Classical and hybrid quantum-classical ML analysis of bias in loan default prediction

Group Members: Tan Dai Ngo (V01073190), Joyce van de Vegte (V00102221), Jeremy Perkins (V00501015), Da Zhang (V01062902), Naomi Phan (V01097400), Yuanming Wang (V01028720)

1. The Problem

The project will examine prediction bias in loan default prediction. Financial institutions that loan money use verified information from client application forms to make loan approval determinations. These determinations are informed by records of whether past clients repaid their loans or defaulted on them. Were a model to misclassify a new client as someone who would default on a loan, a loan application would be inappropriately rejected, with potentially severe consequences for the potential borrower. The converse problem from the bank's perspective is approving a loan that ultimately defaults.

The dataset selected for the project contains 45,000 detailed borrower applications labelled with the loan repayment outcome: default or paid. Features in the dataset include: age, gender, education, income, employment experience, home ownership, requested loan amount, loan intent, interest rate, ratio of loan amount to annual income, credit history, credit score, and previous loan defaults.

The question of bias will pervade every aspect of the project. The exploratory data analysis phase will seek to uncover biases among the features, for example, correlations between income or education with gender, or between credit history and age. Model biases may be learned as a result of imbalances in the dataset, not only by label (loan repaid or not), but also by feature imbalances such as gender, education, or home ownership. A trained model displays bias if its performance metrics vary for demographic subgroups. The reject inference method hypothesizes whether the rejected group would have defaulted. Since the presence or absence of bias in our dataset cannot be known at the outset, backup datasets have been identified to be explored if bias cannot be detected to a sufficient degree in the primary dataset.

Loan repayment datasets:
https://www.kaggle.com/datasets/sumit12100012/loan-approval-classification/data
https://www.kaggle.com/competitions/large-scale-loan-repayment-modeling-challenge/data
Loan approval datasets [1]:
https://www.kaggle.com/datasets/architsharma01/loan-approval-prediction-dataset/data
https://www.kaggle.com/datasets/laotse/credit-risk-dataset/data (plus synthetic version
https://www.kaggle.com/datasets/taweilo/loan-approval-classification-data)

2. Goals

The main goal of this project is to evaluate whether machine learning models can predict loan default/approval accurately. The success will be measured by how well each model balances performance, fairness, and the different costs of misclassification in a loan decision setting.

Our first goal is to build reliable baseline models for loan default prediction. We will evaluate models using metrics including accuracy, precision, recall, F1 score, false positive rate (FPR), false negative rate (FNR), RPC AUC, test error, and empirical test risk. We will pay particular attention to FNR and FPR rather than relying only on overall accuracy. FNR means that a borrower who actually defaults is predicted as nondefaulting, which is costly from the lender's perspective. FPR means that a borrower who would have repaid is predicted as likely to default, which is costly from the borrower's perspective.

Our second goal is to measure and reduce bias in the training models. We will compare model performances across borrower subgroups, such as gender, age group, education level, etc. A model will be considered biased if its error rates differ substantially across those groups. We will then try to see if methods like removing sensitive features, balancing the training data set with Synthetic Minority Oversampling Technique (SMOTE), or applying quantum feature engineering and other preprocessing methods, can reduce these subgroup performance gaps.

Finally, we will compare baseline and other debiased model results after cross-validation and hyperparameter tuning. We will use test error, empirical test risk, and confidence intervals to determine whether performance differences between models are meaningful or just due to random variation. Possible outcomes include finding a debiased dataset to improve fairness with only a small loss in accuracy, finding some debiasing methods that reduce one type of bias while increasing another, or finding that the dataset does not show strong bias effects.

3. **Plan**

In our project, we will be focusing on pre-processing raw data and applying various feature engineering approaches based on two main methods. Specifically, we will explore traditional strategies involving normalization and principal component analysis, as well as quantum-based techniques such as quantum extreme learning machines (QELM). Afterwards, we will be comparing the performances of two machine learning models: XGBoost and Neural Network on the original, debiased, and quantum feature-engineered datasets separately. In order to be able to compare each experiment result consistently, for each experiment, several model design choices will be made, including k-fold cross-validation, hyperparameter tuning, and regularization practices such as feature dropout.

After completing the exploratory data analysis, data pre-processing, and dataset balancing stages by the end of June, we estimate that all of the experiments mentioned above will take between July 1st and July 18th to implement, fine-tune, and debug. This timeline would reasonably leave the last part of our project, the performance evaluation to be finished from July 19 to July 25, and then do one or two presentation rehearsals before the first project presentations in class on July 27th.

There will be a wide variety of resources that we will utilize to make sure our project progresses smoothly. For the model training aspect, we will be relying on lab exercises from the course (Random Forest, Neural Networks) and multiple websites for model implementations and configurations (Random Forest [9], XGBoost [8, 10, 11], Neural Network [12]). For data pre-processing and quantum-based feature engineering, we will study research papers [2, 3, 4], and reach out to professors experienced in the field of quantum computing, namely University of Victoria's Dr. Muller and Dr. Stege, for further knowledge.

4. **Task breakdown**

A Gantt chart showing task assignments is provided in Appendix A.

(a) Exploratory Data Analysis

Exploratory Data Analysis (EDA) is a method often used to understand the statistics and underlying distribution of a given dataset. Feature multicollinearity will be explored using a correlation heatmap. Highly correlated features contain very similar information, forcing the model to process redundant data. However, complex models such as neural networks and gradient boosting are robust to feature redundancy and may utilize these correlated features to uncover hidden, non-linear patterns.

Outliers typically arise from two distinct sources: (1) a valid, historically rare event, or (2) erroneous data collection practice, such as misinformation or a typo. Valid outliers contain crucial predictive signals, revealing the edge cases that often challenge traditional models. Conversely, erroneous outliers can heavily distort the decision boundaries, causing overfitting or misleading the model in an entirely different direction. Hence, we attempt to identify all outliers using Isolation Forest - an unsupervised machine learning technique designed for outlier detection using an ensemble of isolation trees. Following detection, we will discard inherently bad records, while valid minority-class edge cases will be amplified via sample weight adjustment or synthetic data injection to ensure the model can sufficiently capture them. However, it is worth noting that reliably separating these two types of outliers usually requires financial domain knowledge and real-world context. Hence, more research is needed at this stage to identify a trustworthy method.

After all "bad" data and features are removed, if any, box plots (for numerical features) and bar charts (for categorical features) will be used to examine the statistics and distributions of all features, identifying which features are highly skewed and/or correlated with the label. Our group also attempts to use Logistic Regression as a base model to understand how each feature influences the prediction, identifying any feature bias. If a feature is highly correlated to a prediction, the model might resort to simply classifying based on whether that feature exists rather than learning the true underlying pattern of the problem. Such a model would generalize poorly in production.

(b) Data Pre-Processing

Following the results that we received from the exploratory data analysis (EDA) stage, we will consider multiple methods to ensure data integrity before utilizing them for model training. Currently, before having access to the actual EDA outcomes, for the data pre-processing phase, we are considering handling missing values either by imputing (mean) or dropping (entire row) them, applying scaling or normalization techniques on the entire dataset depending on the machine learning models, implementing one-hot encoding for categorical features, potentially removing highly correlated features (based on the generated heatmaps in the EDA), and utilizing principal component analysis (PCA), and quantum extreme learning machines (QELM) as feature engineering approaches.

Post-variational Quantum Extreme Learning Machines (QELM) [2][3][4] are relatively new techniques that can be used as a feature-engineering front end to a classical machine learning algorithm. Once expressed as n qubits, the features occupy 2^n dimensional space, from which the most "probable" n-bit measurement is selected. QELM encodes data features as qubit rotations. A distribution of measurements can serve as input to a classical learning model. QELM are purported to be very sensitive to detecting patterns in data, which may mean that QELM pre-processing may amplify biases in the features and make them easier to identify. Qiskit and Pennylane are open-source Python libraries for quantum computing and simulation. However, it should be noted that the above-mentioned techniques are subject to change based on the insights from the EDA process.

(c) Dataset Balancing

Balancing a dataset adjusts the distributions of its features to discourage a model from making predictions that favour a majority group; a balanced dataset makes it more difficult for a model to learn biases. SMOTE [5], which creates synthetic examples from a minority group, is one approach that will be used to debias the dataset. This is crucially applied only to the training data and not the test data, since the latter needs to be grounded in reality. Weighting the loss function to prevent misclassifications of minority groups during model training will also be explored.

(d) Model Training

The training dataset consists of a list of people who have already been approved for loans, with labels that indicate whether they paid or defaulted on those loans. The dataset itself is biased in terms of who is included then: If there are certain minority groups who are not being granted loans as often, they will be under-represented. This effect will cause a model to learn that minorities are only able to pay loans if they have much higher incomes, for example. Since lower-income minorities may be underrepresented, the model will tend to predict that they will default across the board. In credit scoring, this is referred to as reject inference [6][7], where underrepresented data is missing-not-at-random (MNAR). While previous research has used a variety of solutions to overcome this problem, outlined in [7], we apply SMOTE to insert data into the training set only. Importantly, this involves assumptions about the missing data: We will infer that the censored population shares a similar underlying distribution among the features, and this may not be the case. Therefore, our model essentially tests whether this assumption is correct or not. Its failure would signal that the underlying assumptions need to change.

Regarding the model itself, we will employ gradient boosting, following the suggestions in [7]. While LightGBM was recommended by [7], its advantages seem to be mainly in dealing with huge datasets. It also runs a larger risk of overfitting to noise. Therefore, since our dataset is not massive, we plan to use XGBoost [8], which is a widely used industry standard that covers both continuous and categorical features, which our dataset has. A feedforward neural network will also be trained and compared as a benchmark for comparison.

The training procedure will involve randomly splitting the dataset into two parts, 80% in the training set and holding 20% out for the final test set. The training data will then have standard 5-fold cross-validation applied for hyperparameter tuning, with each fold used once as a validation fold. Training and testing need to be applied in two separate phases: First on the original dataset and again on the debiased dataset with synthetic data added in.

(e) Performance Evaluation

After hyperparameter tuning using k-fold cross-validation on the training set, the models with selected hyperparameters will be retrained using the full training set, and the selected models will be evaluated on the held-out test set. Model evaluation will proceed in three stages. First, each model will be assessed independently using standard classification matrices, including accuracy, confusion matrix, precision, recall, F1-score, ROC AUC, test error and empirical test risk. These metrics will provide an overall view of predictive performance. Second, tuned baseline models will be compared against tuned debiased models. The comparison will be based on test error, empirical test risks, and Hoeffding confidence intervals with Bonferroni adjustments to account for multiple comparisons. Third, because the project focuses on bias in loan default prediction, model performance will also be evaluated across relevant demographic and socioeconomic subgroups, such as gender, age group, education level, income group, and homeownership status. Subgroup performance matrices will be compared to determine whether the model performs unevenly across groups; this step will be the “biasness-fairness” assessment.

Particular attention will be paid to costly misclassification errors. If the positive class is defined as loan default, false negatives represent defaulting borrowers incorrectly predicted as non-defaulting, which is costly from the lender’s perspective. Conversely, false positives represent borrowers incorrectly predicted to default, which may lead to unfair loan rejection and is especially relevant to borrower fairness. On the one hand, lenders need to protect themselves, therefore valuing FPR more, whereas applicants would prefer FNR. We will do some more research in order to arrive at the final cost matrix.

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Appendix A: Task Breakdown and Timeline

	Task Breakdown	Owner
EDA	*Detect and handle outliers using Isolation Forest*	Dai, Jeremy
	Explore data statistics	Naomi
	Evaluate feature importance	Jeremy
	Research QELM	Joyce, Naomi
Data Pre-Processing	*Handle missing values*	Yuanming
	Scale/Normalize data	Yuanming
	One-hot encoding	Yuanming
	Handle highly-correlated features	Naomi
	Apply PCA	Joyce
	Apply QELM	Joyce, Naomi
Dataset Balancing	*Apply SMOTE (class imbalance)*	Dai, Adam, Yuanming
	Sample weight adjustments (outliers)	Adam
Model Training	*XGBoost*	Adam, Naomi, Dai
	Neural Network	Joyce, Jeremy, Yuanming
	K-fold Cross Validation	Everyone
	Hyperparameter tuning	Everyone
Evaluation	*Standard metrices*	Everyone
	Baseline vs. Debiased vs. Quantum-enhanced models	Everyone
	Detect bias towards minority group	Everyone
Report	*Presentation and Written Report*	Everyone

